

Routh-Hurwitz Stability Criterion

Counting right-half-plane roots without solving for them.

Ogata, *Modern Control Engineering*, Ch. 5.

In this tutorial you will:

- build a Routh array and read off the number of unstable roots,
- find the range of a gain K that keeps a loop stable, and
- confirm the boundary against the actual closed-loop poles.

Step through with **Ctrl+Enter**, or render a report with **publish**.

A linear system is stable if and only if all poles of its closed-loop transfer function lie in the open left half of the s -plane. The Routh-Hurwitz criterion determines how many roots of a polynomial

$$a_0 s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n = 0$$

lie in the right half-plane **without solving for the roots explicitly** – useful both as a hand-calculation tool and, as shown below, for finding a symbolic range of a design parameter (e.g. a controller gain) for which a closed-loop system remains stable.

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Necessary Condition

A necessary (but not sufficient) condition for stability is that all coefficients $a_0, \dots, a_n$ be present and of the same sign. If any coefficient is zero or negative (with others positive), the system is immediately known to be unstable – no further test is needed.

Routh Array Construction

For $a_0 s^4 + a_1 s^3 + a_2 s^2 + a_3 s + a_4 = 0$, the array is

```
s\^{4} \ensuremath{[]}\ a0\ a2\ a4
s\^{3} \ensuremath{[]}\ a1\ a3\ 0
s\^{2} \ensuremath{[]}\ b1\ b2
s\^{1} \ensuremath{[]}\ c1
s\^{0} \ensuremath{[]}\ d1
```

with

$$b_1 = \frac{a_1 a_2 - a_0 a_3}{a_1}, \quad b_2 = \frac{a_1 a_4 - a_0 \cdot 0}{a_1} = a_4$$

$$c_1 = \frac{b_1 a_3 - a_1 b_2}{b_1}, \quad d_1 = \frac{c_1 b_2 - b_1 \cdot 0}{c_1} = b_2$$

The number of sign changes in the first column equals the number of roots in the right half-plane; the system is stable iff there are no sign changes (all first-column entries strictly positive, given $a_0 > 0$).

Worked Example: Numeric Routh Array

$$s^4 + 2s^3 + 3s^2 + 4s + 5 = 0$$

```
a = [1 2 3 4 5];
a0=a(1); a1=a(2); a2=a(3); a3=a(4); a4=a(5);

b1 = (a1*a2-a0*a3)/a1;
b2 = a4;
c1 = (b1*a3-a1*b2)/b1;
d1 = b2;

first_col = [a0 a1 b1 c1 d1];
fprintf('Routh first column: %s\n', mat2str(first_col,4))
sign_changes = sum(diff(sign(first_col)) ~= 0);
fprintf('Sign changes = %d -> %d right-half-plane root(s)\n', sign_changes, sign_changes)

% Verify against the actual roots
r = roots(a);
disp('Roots:'); disp(r)
disp('Number with positive real part:'); disp(sum(real(r)>0))

Routh first column: [1 2 1 -6 5]
Sign changes = 2 -> 2 right-half-plane root(s)
Roots:
-1.2878 + 0.8579i
-1.2878 - 0.8579i
0.2878 + 1.4161i
0.2878 - 1.4161i

Number with positive real part:
2
```

Worked Example: Stability Range for a Gain Parameter

Consider a unity-feedback system with forward path

$$G(s) = \frac{K}{s(s+1)(s+2)}$$

The closed-loop characteristic equation is

$$1 + G(s) = 0 \Rightarrow s(s+1)(s+2) + K = 0 \Rightarrow s^3 + 3s^2 + 2s + K = 0$$

Routh array:

```
s^3 \ensuremath{||} 1 2
s^2 \ensuremath{||} 3 K
s^1 \ensuremath{||} (6-K)/3
s^0 \ensuremath{||} K
```

Stability requires every first-column entry positive: $3 > 0$ (always), $(6 - K)/3 > 0 \Rightarrow K < 6$, and $K > 0$. Hence

$$0 < K < 6$$

```
syms s K
charpoly = s*(s+1)*(s+2) + K;
charpoly = expand(charpoly)

% Numerically sweep K and confirm the closed-loop poles cross into the
% right half-plane exactly at the predicted boundary K = 6.
Ks = [0.1 1 3 5.9 6 6.1 8];
for k = Ks
    p = roots([1 3 2 k]);
    fprintf('K = %5.2f -> max real part of poles = %7.4f\n', k, max(real(p)))
end
```

```

charpoly =

s^3 + 3*s^2 + 2*s + K

K = 0.10 -> max real part of poles = -0.0544
K = 1.00 -> max real part of poles = -0.3376
K = 3.00 -> max real part of poles = -0.1642
K = 5.90 -> max real part of poles = -0.0046
K = 6.00 -> max real part of poles = 0.0000
K = 6.10 -> max real part of poles = 0.0045
K = 8.00 -> max real part of poles = 0.0832

```

Special Case: Zero in the First Column

If a first-column entry is zero (but the row is not entirely zero), replace it with a small positive constant ϵ and continue; the sign of the limit as $\epsilon \rightarrow 0^+$ reveals the sign change. If an entire row is zero, it indicates roots symmetric about the origin (e.g. purely imaginary pairs), and the array is continued using the derivative of the auxiliary polynomial formed from the row above. These special cases are not required for the worked examples above, which both have well-defined, nonzero first columns.

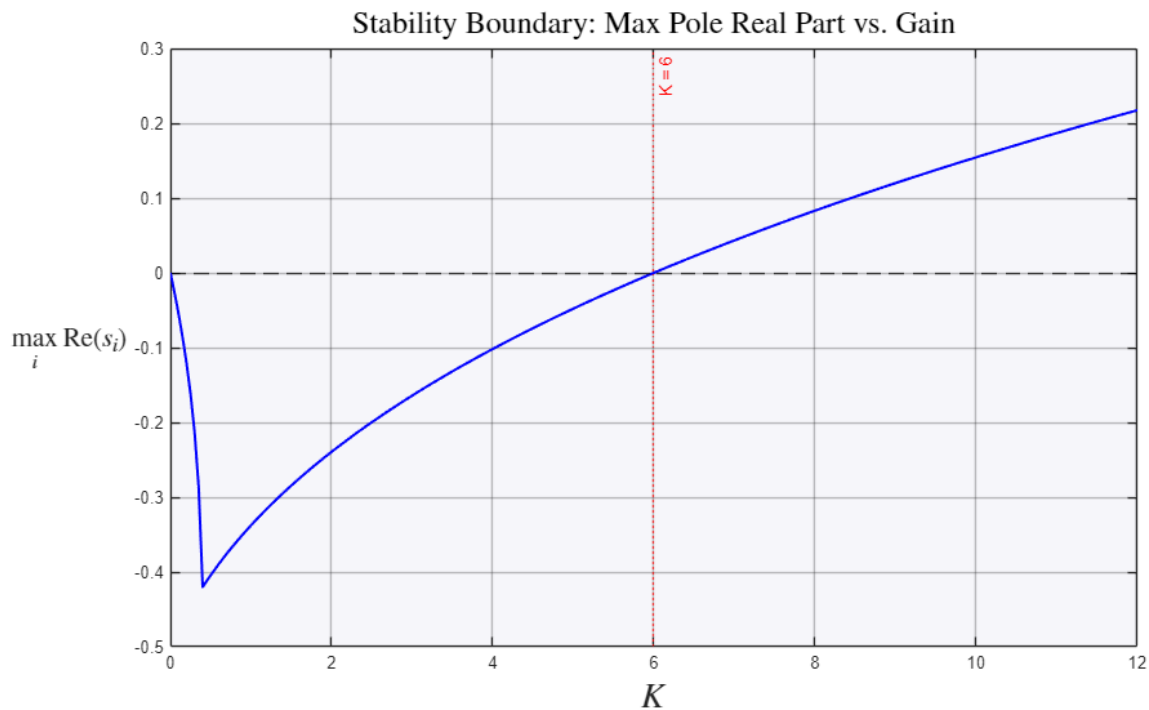
Visualizing the Stability Boundary

Sweeping the gain K of the $K/(s(s+1)(s+2))$ example and plotting the maximum real part of the closed-loop poles shows them crossing into the right half-plane exactly at $K = 6$ – the boundary the Routh array predicted analytically above.

```

K_sweep = 0:0.05:12;
max_re = zeros(size(K_sweep));
for ii = 1:numel(K_sweep)
    max_re(ii) = max(real(roots([1 3 2 K_sweep(ii)])));
end
figure
plot(K_sweep, max_re, 'b', 'LineWidth', 1.5)
hold on
yline(0, 'k--')
xline(6, 'r:', 'K = 6')
hold off
grid on
title('Stability Boundary: Max Pole Real Part vs. Gain', 'Interpreter', 'latex', 'FontSize', 18)
ylabel('$\max_i \mathrm{Re}(s_i)$', 'Interpreter', 'latex', 'FontSize', 16)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$K$', 'Interpreter', 'latex', 'FontSize', 20)

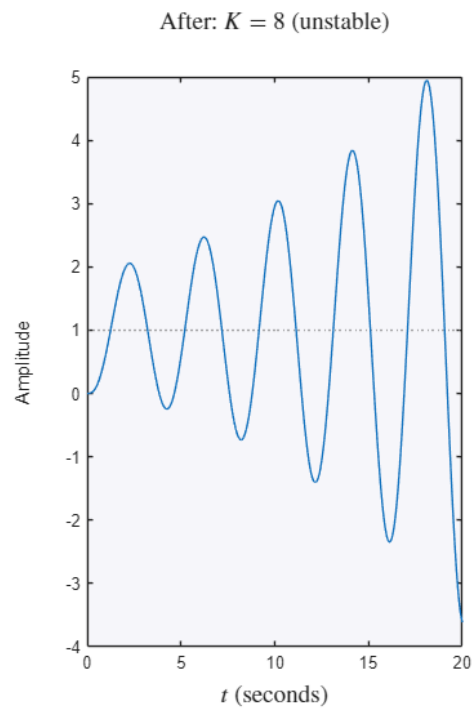
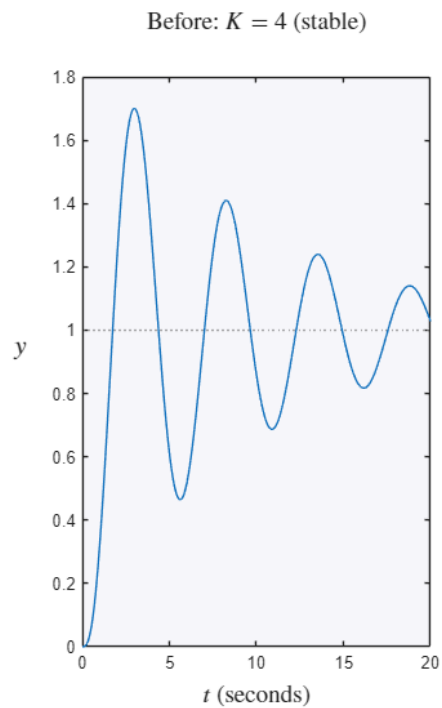
```



Before vs. After: Stable vs. Unstable Gain

The boundary in the time domain: at $K=4$ (< 6) the closed-loop step settles; at $K=8$ (> 6) it diverges – crossing $K=6$ flips stability.

```
t_cl = 0:0.01:20;
figure
subplot(1,2,1)
step(feedback(tf(4,[1 3 2 0]),1), t_cl)
title('Before: $K=4$ (stable)', 'Interpreter', 'latex', 'FontSize', 14)
ylabel('$y$', 'Interpreter', 'latex', 'FontSize', 16)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$', 'Interpreter', 'latex', 'FontSize', 14)
subplot(1,2,2)
step(feedback(tf(8,[1 3 2 0]),1), t_cl)
title('After: $K=8$ (unstable)', 'Interpreter', 'latex', 'FontSize', 14)
xlabel('$t$', 'Interpreter', 'latex', 'FontSize', 14)
```



Try it yourself

- Extend the plant to $s(s+1)(s+2)(s+3)$ (fourth order), rebuild the Routh array, and notice the stability range for K shrink.
- Set K exactly at the boundary ($K=6$) and confirm `roots` shows a pair on the imaginary axis – marginal stability.